Welcome to
Total Quality Management
(19MEE347)

Unit 1 – Quality





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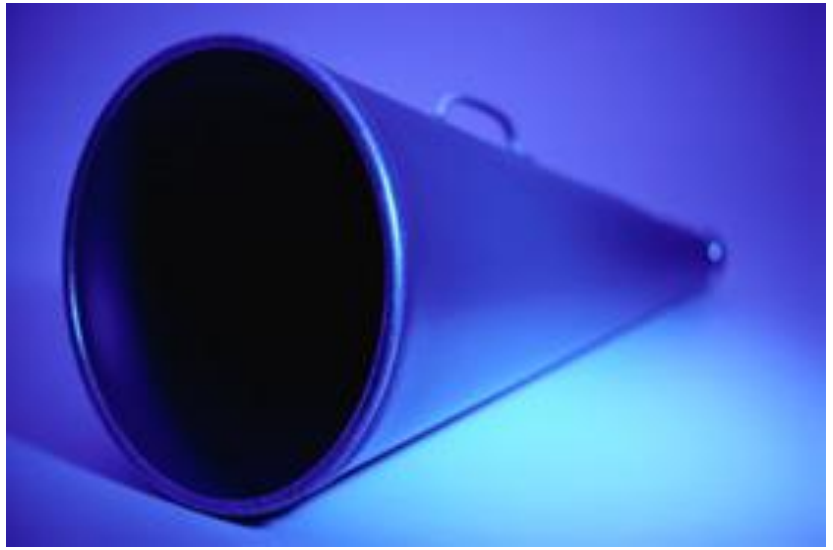
Introduction



Be on time



Focus on the issues



Keep confidences



Participate positively

Introduction to TQM

TQM: The word 'total' implies that all the members of the organization make consistent efforts to achieve the objectives through systematic efforts for improvement of the organization *(through customer satisfaction)*.

- Customer oriented management
- Customer satisfaction
- Customer needs and expectations
- Customer delight





What is QUALITY?

- **What is quality?**

  "meeting the customer requirements"

- **What is reliability?**

  "It is the ability of product and service to **continue to meet the customer requirements**"

- **Quality is meeting the customer requirements, and this is not restricted to the functional characteristics of the product or services.**





Meaning of Quality: Consumer's Perspective

- **Fitness for use**
  - How well product or service does it is supposed to
- **Quality of design**
  - Designing quality characteristics into a product or service
  - A Mercedes and a Ford are equally "fit for use," but with different design dimensions





Definition of Quality

- Quality is also defined as **excellence** in the product or service that **fulfills or exceeds the expectations of the customer.**
- Though quality is an abstract perception, it has a quantitative measure - **Q = (P / E),**

  where,

  Q = quality,

  P = performance (as measured by the Mfgr.), and

  E = expectations (of the customer).





Definition of Quality

- Quality is **not fine-tuning your product at the final stage** of manufacturing, before packaging and shipping.
- **Quality** is **judgment by customers.** It is customer determination, not an engineer's determination or a manager's determination.

QUALITY AS FOLLOWS

- Quality is **total composite product** (goods and services) **characteristics**, through which the product in use will meet the needs and expectations of the customers.
- **Concept of quality** must start **with identification of customer quality requirements** and must end only when the finished product is placed into the hands of the customer who **remains satisfied** through various stages of relationship with the seller.

Definition of Quality

- **American Society of Quality Control (ASQC) and American National Standard Institute (ANSI)** defined
  - Quality is totality of features and characteristics of product (goods and services) that bears on its ability to satisfy given needs"

- **ISO 9000:2000 -** Quality is the degree to which a set of inherent characteristics fulfills requirements.
- Deming – Quality is a predictable degree of uniformity and dependability, at low cost and suited to the market.
- Joseph M. Juran - Quality is fitness for use or purpose.
- Philip. B. Crosby – Quality is conformance to requirements.





Approaches to define Quality

- **Transcendent Approach**
  - Quality is absolute and universally recognisable.
  - It is common notion used by laymen
  - There is no subjective judgement and is estimated by looking at the product
- **Product Based Approach**
  - Attributes of a particular product in a specific category
  - These attributes are accepted as bench of quality by the industry
  - Others in the same industry try to produce close to this quality





Approaches to define Quality

- **User Based Approach**
  - Defined as "Fitness for use" (conform to essential requirements)
  - Viewed from user's perspective and is dependent on how well does the product meet the needs of the consumer.
  - Also known as Customer Oriented Approach
- **Production Based Approach**
  - An outcome of engineering or operational excellence and is measured in terms of quality of conformance.
  - The producer has specifications and produces the product as per the specifications.





Approaches to define Quality

- **Value Based Approach**
  - Quality is viewed in context of price
  - Quality is satisfactory, if it provides desired performance at an acceptable price
  - Customer looks at the total value proposition and not the price alone.

$$Value = \frac{Benefits}{\Pr ice}$$





Dimensions of Quality

Manufacturing or product Dimensions

- Performance
- Features
- Reliability
- Conformance (Standards of Something)
- Durability
- Serviceability
- Responsiveness
- Aesthetics
- Reputation

Service Dimensions

- Time and Timeliness
- Completeness
- Courtesy
- Consistency
- Accessibility and convenience
- Accuracy
- Responsiveness





Dimensions of Quality: Manufactured Products

- *Performance* - main or primary characteristics of the product/service (Eg. Audio quality of the mobile phone)
- *Features* – Secondary characteristics (Eg. Face book facility, camera and music system)
- *Reliability* - consistency of performance. It is a measure of **mean time between failures (MTBF)**. (Eg. The MTBF of a mobile phone can be specified in No. of months)
- *Conformance* - how well product/service **conforms to the standards** (national, international and industry) (Eg. Some mobile phones are banned in India as they fail to conform the government regulations)





Dimensions of Quality: Manufactured Products

- *Durability* - useful life of the product/service (Eg. The durability of a low price mobile phone can be considered as 2 to 3 years)
- *Serviceability* - service after sale or repair a product quickly and easily. (Eg. Faster mobile service providers and economical service)
- *Responsiveness* - It is a measure of how well the manufacturer of the product is able to adapt to the changing needs of the customer. (Eg. Dual sim or multi sim to satisfy the customer need)
- *Aesthetics* - appearance, feel, smell, taste and colour. (Eg. Mobile model, slim in size, touch screen)
- *Reputation* – Brand name or perceived quality (Eg. Apple and Samsung are branded mobile manufactures in India)





Dimensions of Quality: Service

- **Time and Timeliness**
  - How long must a customer wait for service, and is it completed on time?
  - Is an overnight package delivered overnight?
- **Completeness**
  - Is everything customer asked for provided?
  - Is a mail order from a catalogue company complete when delivered?
- **Courtesy**
  - How are customers treated by employees?
  - Are catalogue phone operators nice and are their voices pleasant?
- **Consistency**
  - Is the same level of service provided to each customer each time?
  - Is your newspaper delivered on time every morning?



Dimensions of Quality: Service

- **Accessibility and convenience**
  - How easy is it to obtain service?
  - Does a service representative answer you calls quickly?
- **Accuracy**
  - Is the service performed right every time?
  - Is your bank or credit card statement correct every month?
- **Responsiveness**
  - How well does the company react to unusual situations?
  - How well is a telephone operator able to respond to a customer's questions?





Meaning of Quality: Producer's Perspective

- **Quality of Conformance**
  - Making sure a product or service is produced according to design
    - if new tires do not conform to specifications, they wobble
    - if a hotel room is not clean when a guest checks in, the hotel is not functioning according to specifications of its design





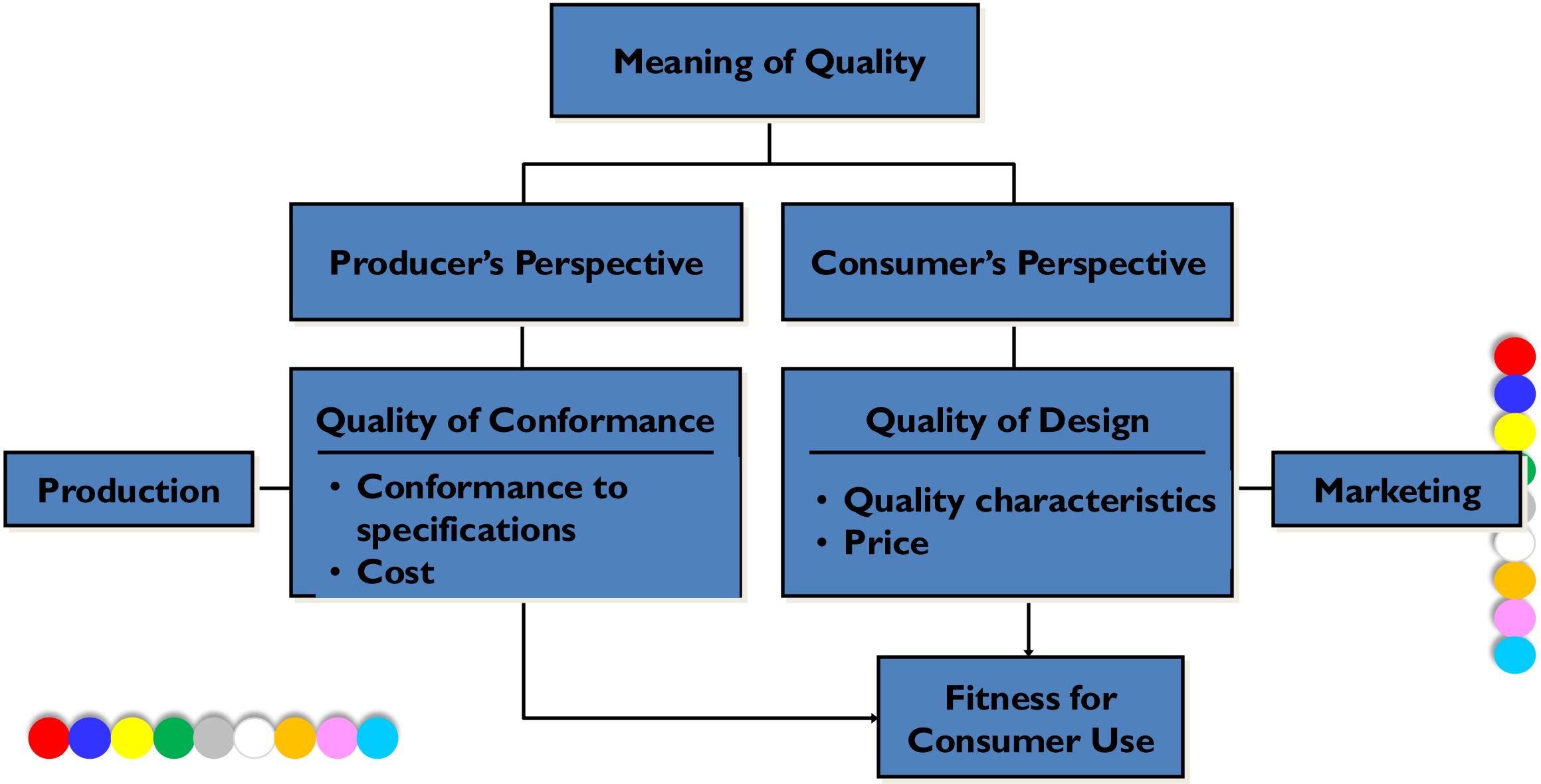
Meaning of Quality: A Final Perspective

- Consumer's and producer's perspectives depend on each other
- Consumer's perspective: PRICE
- Producer's perspective: COST
- Consumer's view must dominate





Meaning of Quality



Three Spheres of Quality



Three Spheres of Quality

- **Quality Control**
  - Includes phases of analysis, relation, and generalization.
- **Activities relating to quality control include:**
  - Monitoring process capability and stability
  - Measuring process performance
  - Reducing process variability
  - Optimizing processes to nominal measures
  - Performing acceptance sampling
  - Developing and maintaining control charts





Three Spheres of Quality

- **Quality Assurance**
  - Refers to activities associated with guaranteeing the quality of a product or service.
- **Quality assurance activities include tasks such as:**
  - Failure mode and effects analysis
  - Concurrent engineering (a method of designing and developing products, in which the different stages run simultaneously, rather than consecutively)
  - Experimental design
  - Process improvements
  - Design team formation and management
  - Off-line experimentation
  - Reliability/durability product testing





Three Spheres of Quality

- **Quality Management**
  - The management processes that overarch and tie together the control and assurance activities.
- **Quality management activities:**
  - Planning for quality improvement.
  - Creating a quality organizational culture.
  - Providing leadership and support.
  - Providing training and retraining.
  - Designing an organizational system that reinforces quality ideals.
  - Providing employee recognition.
  - Facilitating organizational communication.





Quality Policy

- Quality policy Statement serves as a **guide for everyone** in the organization.
- This statement clarifies the employees about **how the products and services must be provided to the customers.**
- The CEO of the company writes quality Policy Statement after a careful study and analysis of the feedback from the workforce.
- Finally, the **Quality Council must approve the statement.**





Quality Policy

In establishing the quality policy, top management should consider

— the level and type of future improvement needed for the organization to be successful,

— the expected or desired degree of customer satisfaction,

— the development of people in the organization,

— the needs and expectations of other interested parties,

— the resources needed to go beyond ISO 9001 requirements, and

— the potential contributions of suppliers and partners





Quality Policy

Top Management shall ensure that the Quality Policy is:

- Is appropriate to the purpose of the organization.
- Include a commitment to comply with requirements and continually improve the effectiveness of the quality management system.
- Frame work for establishing and reviewing the effectiveness of quality objectiveness.
- IS communicated and understood within the organization.
- And reviewed for **continuous suitability**.





Quality Management Processes

- Quality Planning
- Quality Assurance
- Quality Control





Quality Planning

- The process of identifying which quality standards are relevant to the project and determining how to satisfy them.
- ***Input includes***: *Quality policy (Commitment to customer requirement), scope statement, product description, standards and regulations, and other process Output.*
- ***Methods used***: *benefit / cost analysis, benchmarking, flowcharting, and design of experiments*
- ***Output includes***: *Quality Management Plan (setting clear quality standards, defining roles and responsibilities, and outlining), operational definitions, checklists, and Input to other processes.*





Quality Assurance

- ***Quality assurance is about engineering "processes" that assure quality.***
- The process of evaluating overall project performance on a regular basis to provide confidence that the project will satisfy the relevant quality standards.
- **Input includes:** Quality Management Plan, results of quality control measurements, and operational definitions.
- **Methods used:** quality planning tools and techniques and quality audits.
- **Output includes**: quality improvement.





Quality Assurance

- **Quality Assurance (QA)** is a **process driven approach**, it is a process to monitor and improve existing quality processes.
- It is a process of verifying whether the software product or services meets or exceeds the customer expectations.
- It ensures that the product or services are developed or implemented on agreed standards.
- Quality assurance ensures that the **processes designed for the product development and services** are **effective enough to meet the objectives.**
- It prevents software defects/errors.





Quality Assurance

- ***Testing might help in detecting defects, but not in avoiding them.***
- A **defect once fixed cannot ensure that it won't occur again**, even **if the root cause is found**. The **process or the system** that allowed that defect to occur is what **needs to be re-engineered**, and this is what is called **quality assurance**.
- Everyone who is involved in the end to end development process, including analysts, developers, testers, managers etc., is an important player in assuring quality.
- In fact, **Quality Assurance might not involve testing at all**.





Quality Control

- The process of monitoring specific project results to determine if they comply with relevant quality standards and identifying ways to eliminate causes of unsatisfactory performance.
- ***Input includes****: work results, Quality Management Plan, operational definitions, and checklists.*
- ***Methods used include****: inspection, control charts, pareto diagrams, statistical sampling, flowcharting, and trend analysis.*
- ***Output includes****: quality improvements, acceptance decisions, rework, completed checklists, and process adjustments*





Quality Control

- **Quality Control (QC)** - "the operational techniques and activities that are used to fulfil requirements for quality".
- The purpose of quality control is to uncover defects and have them corrected so that defect-free products will be produced.
- Quality control is limited to looking at *products* .
- Quality control is testing the final product against product quality standards.

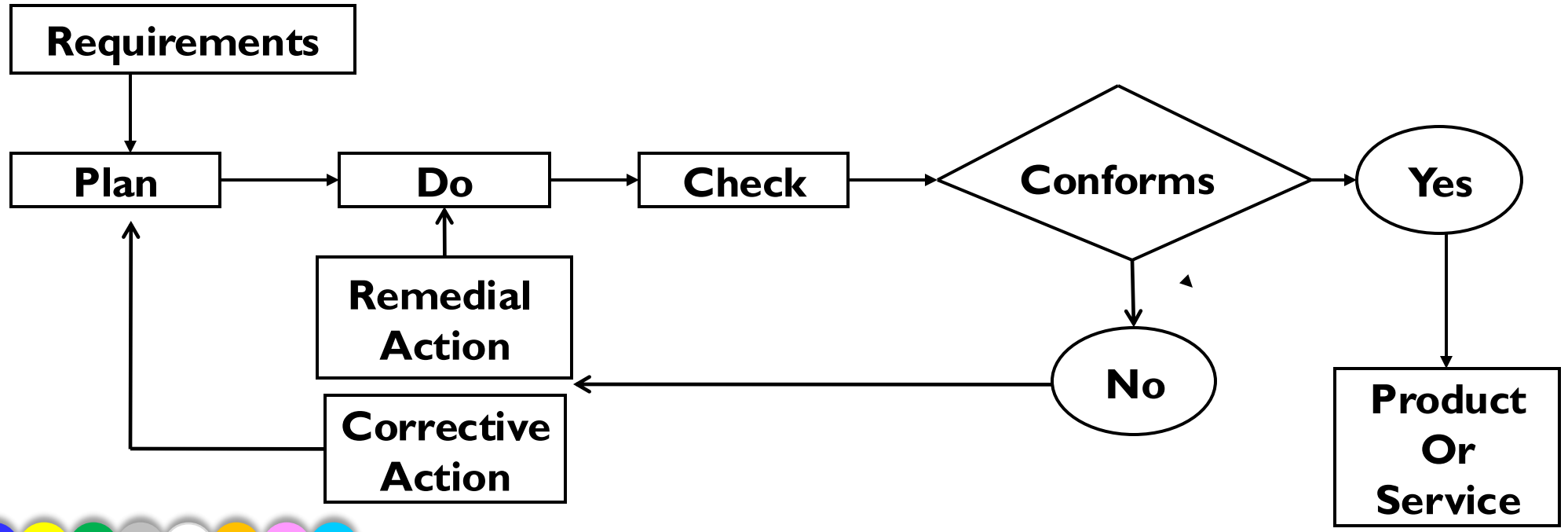




Quality Control

It is that part of Quality Management focused on fulfilling requirements of the Customers for the quality products.

A Simplest Form of Quality Control is:-



Cost of Quality

- **Cost of Achieving Good Quality**
  - **Prevention costs**
    - costs incurred during product design
  - **Appraisal costs**
    - costs of measuring, testing, and analyzing
- **Cost of Poor Quality**
  - **Internal failure costs**
    - include scrap, rework, process failure, downtime, and price reductions
  - **External failure costs**
    - include complaints, returns, warranty claims, liability, and lost sales





Prevention Costs

- ***Quality planning costs***
  - costs of developing and implementing quality management program
- ***Product-design costs***
  - costs of designing products with quality characteristics
- ***Process costs***
  - costs expended to make sure productive process conforms to quality specifications
- ***Training costs***
  - costs of developing and putting on quality training programs for employees and management
- ***Information costs***
  - costs of acquiring and maintaining data related to quality, and development of reports on quality performance





Appraisal Costs

- ***Inspection and testing***
  - costs of testing and inspecting materials, parts, and product at various stages and at the end of a process
- ***Test equipment costs***
  - costs of maintaining equipment used in testing quality characteristics of products
- ***Operator costs***
  - costs of time spent by operators to gather data for testing product quality, to make equipment adjustments to maintain quality, and to stop work to assess quality





Internal Failure Costs

- ***Scrap costs***
  - costs of poor-quality products that must be discarded, including labor, material, and indirect costs
- ***Rework costs***
  - costs of fixing defective products to conform to quality specifications
- ***Process failure costs***
  - costs of determining why production process is producing poor-quality products
- ***Process downtime costs***
  - costs of shutting down productive process to fix problem
- ***Price-downgrading costs***
  - costs of discounting poor-quality products—that is, selling products as "seconds"

External Failure Costs

- **Customer complaint costs**
  - costs of investigating and satisfactorily responding to a customer complaint resulting from a poor-quality product.
- **Product return costs**
  - costs of handling and replacing poor-quality products returned by customer.
- **Warranty claims costs**
  - costs of complying with product warranties.
- **Product liability costs**
  - litigation costs resulting from product liability and customer injury
- **Lost sales costs**
  - costs incurred because customers are unsatisfied with poor quality products and do not make additional purchases





Measuring and Reporting Quality Costs

- **Index numbers**
  - ratios that measure quality costs against a base value
  - **labor index**
    - ratio of quality cost to labor hours
  - **cost index**
    - ratio of quality cost to manufacturing cost
  - **sales index**
    - ratio of quality cost to sales
  - **production index**
    - ratio of quality cost to units of final product





TQM: Historical review

- Quality in articles and artefacts produced by skilled craftsmen and artisans from the B.C. era. Eg. goldsmiths, silversmiths, blacksmiths, potters, etc.
- Artists & Artisans Guilds in the Middle ages **spent years imparting quality skills** and the worksmen had pride in making quality products.
- Industrial Revolution brought factory manufacturing where articles were mass-produced and each worker made only a part of the product, and did not sense the importance of his contribution to the quality of the product.





TQM: Historical review

- In 1924, W. A. Shewhart of Bell Telephone Labs developed a statistical chart for the control of product variables – the beginning of SQC and SPC.
- In the same decade, H. F. Dodge and H. G. Romig of Bell Telephone Labs developed statistical acceptance sampling instead of 100% inspection.
- In 1946, the American Society for Quality Control was formed.
- In 1950, W. Edwards Deming, who learnt SQC from Shewhart, taught SPC & SQC to Japanese engineers and CEO's.





TQM: Historical review

- In 1954, Joseph M. Juran taught Japanese managements their responsibility to achieve quality.
- In 1960, the first quality control circles were formed. SQC techniques were being applied by Japanese workers.
- 1970's US managers were learning from Japan Quality implementation miracles.
- In 1980's TQM principles and methods became popular. (also in auto industry)
- In 1990's, the ISO 9000 model became the world-wide standard for QMS.





Quality Philosophies

- Edward Deming
- Philip Crosby
- Joseph Juran



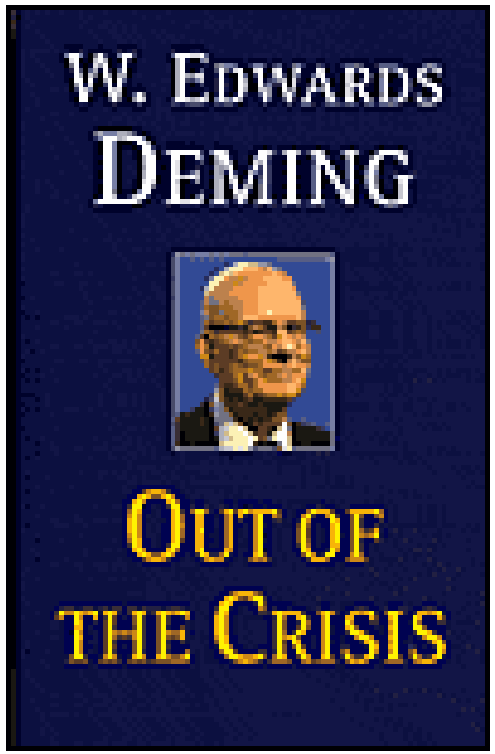


Quality Philosophies

Writer	Definition of Quality	Orientation	Developed
Deming	Fitness for Purpose	Customer	• Deming's Philosophy • 14 Points of Quality • PDSA Cycle • System of Profound knowledge
Juran	Fitness for use	Customer	• Quality Trilogy • Five Quality Characteristics, • Quality Council, The four phases of Problem solving, The quality Spiral
Crosby	Conformance to requirements	Supplier	• Five absolutes of quality, • Fourteen point plan for quality

Quality Philosophies

The Quality Guru – Edward Deming



1986

- Quality is "uniformity and dependability"
- Focus on SPC and statistical tools
- "14 Points" for management
- PDSA method



1900-1993

Quality Philosophies

The Quality Gurus – Edward Deming

Deming was an American engineer, statistician, professor, author, lecturer, and management consultant.

Educated as an electrical engineer and later specializing in mathematical physics.

Deming was a professor of statistics at New York University's graduate school of business administration, and taught at Columbia University's graduate school of business.

He also was a consultant for private business.





Quality Philosophies

Deming's Philosophy

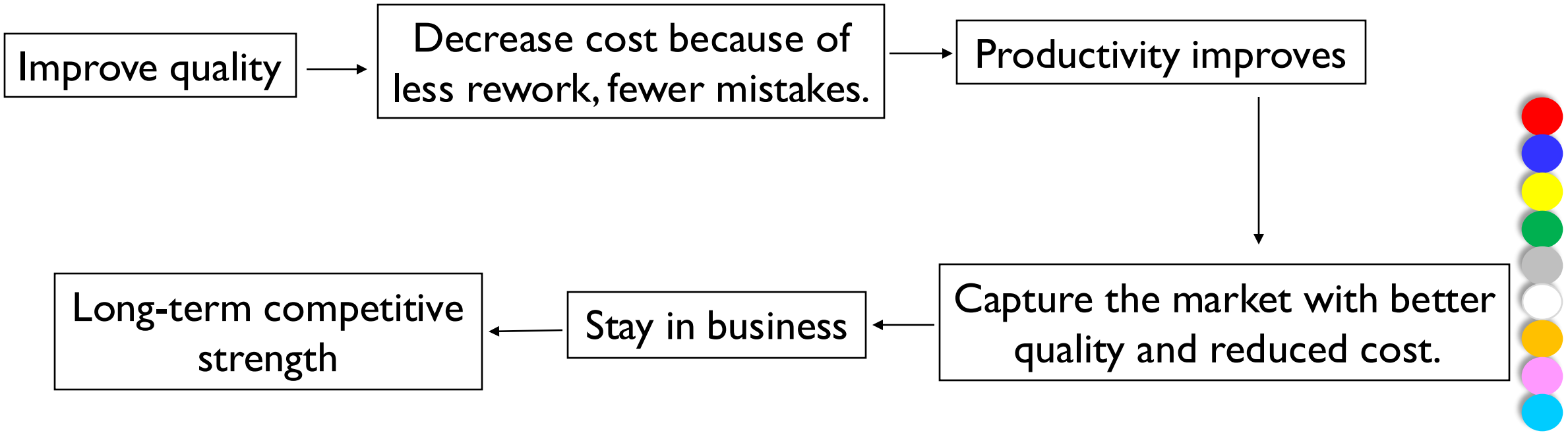
"A System of Profound Knowledge"

1. Appreciation for a system - A system is a set of functions or activities within an organization that work together to achieve organizational goals. Management's job is to *optimize the system*. (not parts of system, but the whole!). System requires *co-operation*.
2. Psychology – The designers and implementers of decisions are *people*. Hence understanding their psychology is important.
3. Understanding process variation – A production process contains many sources of variation. *Reduction in variation improves quality*. Two types of variations- *common causes and special causes*. Focus on the special causes. Common causes can be reduced only by change of technology.
4. Theory of knowledge – Management decisions should be driven by *facts, data and justifiable theories*. Don't follow the managements fads!

Quality Philosophies

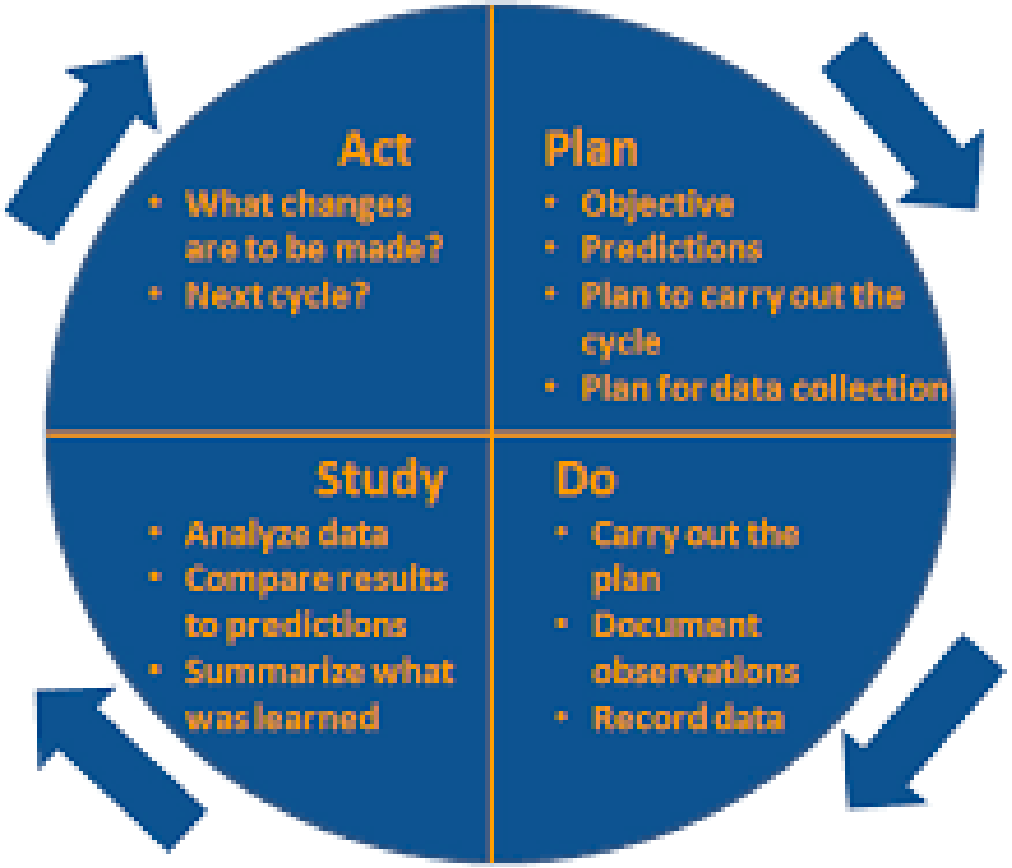
Deming Philosophy

Definition of quality, "**A product or a service possesses quality if it helps somebody and enjoys a good and sustainable market.**"



Quality Philosophies

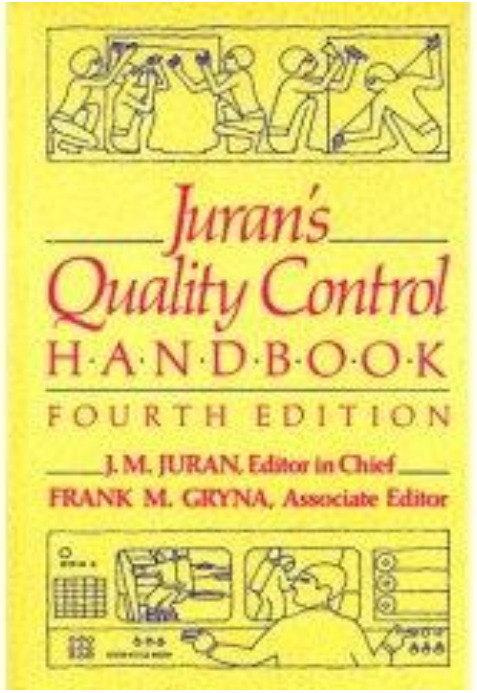
Deming's Advocacy



PDSA Cycle

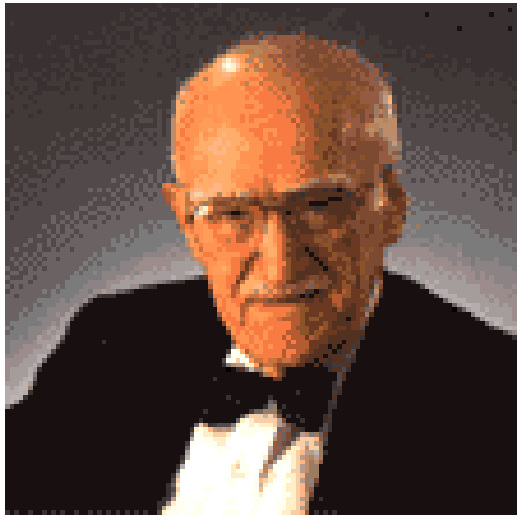
Quality Philosophies

The Quality Guru – Joseph Juran



1951

- Quality is "fitness for use"
- Pareto Principle
- Cost of Quality
- General management approach as well as statistics



1904 - 2008

Quality Philosophies

The Juran philosophy

Pursue quality on two levels:

1. The *mission of the firm* as a whole is to achieve high *product quality*.
2. The *mission of each individual department* is to achieve high *production quality*.

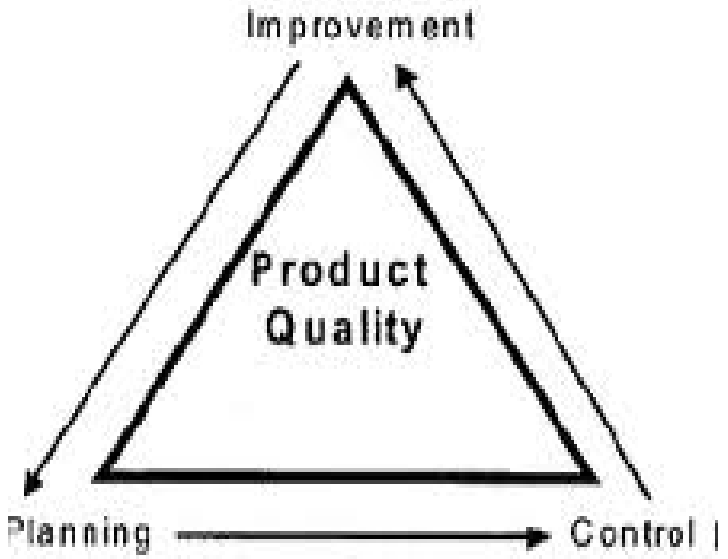
- Quality should be talked about in a language, senior management understands: money (cost of poor quality).
- At operational level, focus should be on conformance to specifications through elimination of defects - use of statistical methods.





Quality Philosophies

The Juran philosophy



Quality Trilogy –

- Product should be a good price,
- Work well for the customer,
- Be distributed efficiently from the producer to the customer, and
- Be supported efficiently by the company

1. *Quality planning*: Process of preparing to meet quality goals. Involves understanding customer needs and developing product features.
2. *Quality control*: Process of meeting quality goals during operations. Control parameters. Measuring the deviation and taking action.
3. *Quality improvement*: Process for breaking through to unprecedented levels of performance. Identify areas of improvement and get the right people to bring about the change.

Quality Philosophies

The Quality Guru – Philip Crosby

"Quality can't be delegated. The who and why must be understood by top management before the what and how can be launched."

"Quality is Free" and "Zero Defects"



Quality Philosophies

Philip Crosby

- Noted Quality Consultant, Lecturer and Author of many books translated into over 10 languages.
- Author of QUALITY IS FREE, 1979 Best Seller
- Recognized Business Philosopher, Innovator and Quality Guru
- Promoted the concept of "Zero Defects" originally at Martin Marietta where he worked 1960's
- Quality attitude, commitment to quality performance, zero defects first time every time
- Defined Quality as CONFORMANCE TO REQUIREMENTS
- Prevention rather than detection and correction.





Quality Philosophies

Philip Crosby

Main Concepts:

- Management must understand the issues and take responsibility to improve
- Management must remove barriers to quality
- Must understand organization and process capabilities
- Management must continually measure quality by measuring the cost of doing things wrong
- Quality Pays and Pays Handsomely, Quality does NOT cost more per conventional wisdom
- Cost of Quality Analysis (COQ = Price of Non-Conformance + Price of Conformance)





Quality Philosophies

Phillip Crosby

- Four Absolutes of Quality
  - Quality is Conformance to Requirements
  - Eliminate Errors before they Occur
  - Do it right the first time
  - The Measurement of Quality is the Price of Non-Conformance





Quality Philosophies

Phillip Crosby

- Stressed the way to manage quality is by prevention, not detection and testing.
- Addresses the need to change perceptions and attitudes about quality.
- Avoid the common attitude that error is inevitable; it is a normal part of business life, and one must cope.
- The ultimate goal of quality improvement is Zero Defects or defect-free products and services.
- Zero Defects are an attitude and commitment to prevention.
- Zero Defects does not mean that the product has to be perfect.



Processes whose primary purpose is to prevent product manufacturing failures.

Quality Philosophies

Phillip Crosby

Prevention begins by:

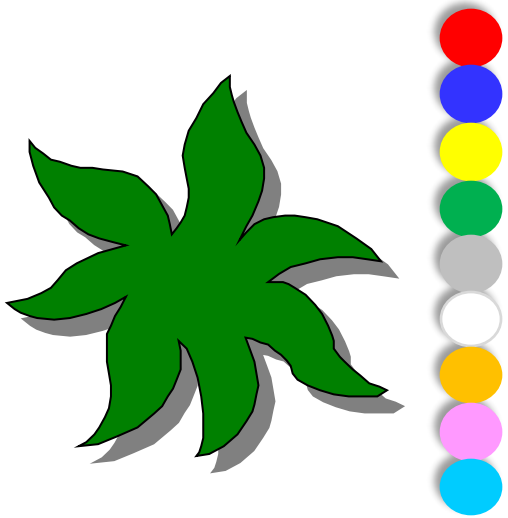
- Establishing product or service requirements,
- Developing the product or service,
- Gathering data, comparing the data to the requirement, and
- Taking action on the result.
- *This is a continuous activity.*





7 Habits of highly effective people (Stephen Covey)

- Be pro-active
- Begin with the end in mind
- Put first things first
- Think win-win
- Seek first to understand, then to be understood
- Synergy
- Sharpen the saw



Role of TQM leaders

- All are responsible for quality improvement especially the senior management & CEO's
- Senior management must practice **Managing by walking around (MBWA).**
- Ensure that the team's decision is in harmony with the quality statements of the organization
- Senior TQM leaders must read TQM literature and attend conferences to be aware of TQM tools and methods
- Senior managers must take part in award and recognition ceremonies for celebrating the quality successes of the organization





Role of TQM leaders

- Coaching others and teaching in TQM seminars
- Senior managers must collaborate with internal, external and suppliers through visits, focus groups, surveys
- They must live and communicate TQM.





TQM implementation

- Begins with Sr. Managers and CEO's
- Timing of the implementation process
- Formation of Quality council
- Union leaders must be involved with TQM plans implementation
- Everyone in the organization needs to be trained in quality awareness and problem solving
- Quality council decides QIP projects.





Quality Council

- The **quality council includes the CEO and Senior managers** of the functional areas – research, manufacturing, finance, sales, and marketing, and one coordinator and a union representative.

- Duties - To develop the Quality statements Eg. Vision, Mission, Quality policy statements, Core values, etc.

- To develop strategic long-term plans and annual quality improvement programs.

- Make a quality training program.





Quality Council

- Monitor the costs of poor quality.
- Determine the performance measures for the organization
- Always find projects that improve the processes and produce customer satisfaction.
- Establish work-group teams and measure their progress.
- Establish and review the recognition and reward system for the TQM system





Quality statements

- **Vision statement** – a short declaration of what the organization hopes to be tomorrow.
- **Mission statement** – a statement of purpose – who we are, who are our customers, what we do, and how we do it.
- **Quality policy** – is a guide for everyone in the organization, how they should provide products and services to the customers.





Strategic Planning

Strategic business planning is similar to strategic quality planning.

7 steps to strategic planning

- Customer needs
- Customer positioning (helps determine where your business stands in the market and how it should be positioned to attract more customers)
- Predict the future
- Gap analysis
- Closing the gap
- Alignment
- Implementation.





Strategic Quality Goals and Objectives

- Goals must be focused
- Goals must be concrete
- Goals must be based on statistical evidence
- Goals must have plan or method with resources
- Goals must have a time-frame
- Goals must be challenging yet achievable





Deming’s 14 Key Philosophies





Deming's 14 Key Philosophies

1 - Constancy of Purpose

Create constancy of purpose for continual improvement of products and service to society, allocating resources to provide for long range needs rather than only short Term profitability, with a plan to become competitive, to stay in business, and to provide jobs.

Developing the organizations goals and philosophy

- Long term view
- Stating the Organization's goals and philosophy
- Self examination – where are we
- Developing a Mission Statement
- Making the Mission Statement a "Living" document





Deming's 14 Key Philosophies

2 - Adopt the new philosophy

We are in a new economic age. We can no longer live with commonly accepted levels of delays, mistakes, defective materials, and defective workmanship. Transformation of Western management style is necessary to halt the continued decline of business and industry.

Understanding the Philosophy of never-Ending Improvement

- Customer satisfaction
- Managing for success instead of failure
- Identify and remove barriers to achieving quality
- Get everyone involved in the quality journey





Deming's 14 Key Philosophies

3 - Cease the need for mass inspection

Eliminate the need for mass inspection as the way of life to achieve quality by building quality into the product in the first place. Require statistical evidence of built in quality in both manufacturing and purchasing functions.

Replacing mass inspection with Never-Ending improvement

- Develop a plan that minimizes the total cost of incoming materials and final product
- Inspect all or none rule
- Commit to examining the process over time





Deming's 14 Key Philosophies

4 - End lowest tender contracts

End the practice of awarding business solely on the basis of price tag. Instead require meaningful measures of quality along with price. Reduce the number of suppliers for the same item by eliminating those that do not qualify with statistical and other evidence of quality. The aim is to minimize total cost, not merely initial cost, by minimizing Variation. This may be achieved by moving toward a single supplier for any one item, on a long term relationship of loyalty and trust.

Changing the philosophy of purchasing

- Price has no meaning without a measure of quality being purchased – do not make cost the sole decision factor
- Move from multiple to single source relationships
- Long term relationship between the vendor and buyer
- The lowest price or bidder means poorer quality





Deming's 14 Key Philosophies

5 - Improve every process

Improve constantly and forever the system of production and service, to improve quality and productivity, and thus constantly decrease costs.

Improving the system

- Management has responsibility for the "system"
- Continual reduction of waste
- Continual improvement in quality in every activity
- Management to define operational definitions/communication
- Use of Control Charts, flow Charts, Check Sheets, Pareto Diagrams, Brainstorming, Fishbone (cause and Effect), Histograms, Scatter Diagrams for managing quality
- Shewhart Cycle – "Plan/Do/Check/Act"





Deming's 14 Key Philosophies

6 - Institute training on the job

Institute modern methods of training on the job for all, including management, to make better use of every employee. New skills are required to keep up with changes in materials, methods, product and service design, machinery, techniques, and service.

Instituting Modern Training Methods

- Training in the organizational philosophy
- On-going integrated approach to an employees growth
- Learn how to perform the job
- All employees should learn Dr. Deming's 14 points
- Realize that training is part of everyone's job
- Use statistical methods to determine workers capability
- Training that offers employees a share in the overall philosophy and goals for the organization





Deming's 14 Key Philosophies

7 - Institute leadership

Institute leadership - the aim of supervision should be to help people, machines and gadgets to do a better job. Supervision of management is in need of overhaul, as well as supervision of production workers.

Supervising Never-Ending improvement

- People are penalized for things beyond their control
- Management should remove causes for system variation
- Create a positive supportive atmosphere
- Eliminate fear and mistrust
- Encourage coaching
- Extract feedback
- Workers have to give new systems a chance





Deming's 14 Key Philosophies

8 - Drive out fear

Encourage effective two way communication and other means to drive out fear throughout the organization so that everybody may work effectively and more productively for the company.

Driving out fear:

- Fear causes stress, emotional problems, and absenteeism
- Caused by feeling powerless and having no control
- Do not use as a motivator get people to work in teams
- Elimination of fear starts at the top
- Open channels of communication
- Interaction with the organization - Training in company goals
- Organize and structure teams
- Reward teamwork, quality, and creativity





Deming's 14 Key Philosophies

9 - Break down barriers

Break down barriers between departments. People in research, design, sales, and production must work as a team, to foresee problems of production and in use that may be encountered with the product or service.

Breaking down organizational barriers:

- Employees roles become functional
- Problems in competition, communication and fear arise
- Staff areas have to work as an integrated whole
- Customer and employee surveys should be done
- Improve communication upwards and downwards
- Eliminate performance appraisals
- Training to reduce barriers





Deming's 14 Key Philosophies

10 - Eliminate exhortations

Eliminate slogans, exhortations, and targets for the work force asking for zero defects and new levels of productivity. Such exhortations only create adversarial relationships, as the bulk of the causes of low quality and low productivity belong to the system and thus lie beyond the power of the work force.

Replacing numerical goals, posters and slogans with Never - Ending improvement

- Change to system to help employees achieve goals
- Identify problems/barriers that are causing goals not to be met and eliminate them – get rid of management by objectives
- Goals must be focused on the company's mission in the future
- Goals must have an organizational purpose and aligned with the job





Deming's 14 Key Philosophies

11 - Eliminate arbitrary numerical targets

Eliminate work standards (quotas) on the factory floor. Substitute leadership. Eliminate management by objective. Eliminate management by numbers, Numerical goals. Substitute aids and helpful leadership in order to achieve continual improvement of quality and productivity.

Replace management by numbers with Never-Ending Improvement

- Quotas and standards focus on quantity not quality
- Replace with statistical methods, leadership and training
- Identify process improvements
- By focusing on quality through the use of statistical methods, management provides a roadmap for never-ending improvement





Deming's 14 Key Philosophies

12 - Permit pride in workmanship

Remove barriers that rob the hourly worker of his right to pride of workmanship.

Promoting pride of workmanship:

- The responsibility of supervisors must be changed from sheer numbers to quality.
- This means, inter alia," abolishment of the annual or merit rating and of management by objectives
- Involve employees at all levels of process improvement
- Operationally define job descriptions
- Meet basic work-related needs of employees
- Supply employees with the proper tools, materials, & methods





Deming's 14 Key Philosophies

13 - Encourage education

Institute a vigorous program of education and self improvement,. What an organization needs is not just good people; it needs people that are improving with education.

Educating and retraining everyone:

- Should develop employees for changes in their current jobs
- In the organization's mission and goals
- Statistical training
- View training as long term for the individual
- In fields related to the employees current job
- The employees personal improvement
- Failure to do this creates loss of resources in the future

Deming's 14 Key Philosophies

14 - Top management commitment to action

Put everybody in the company to work to accomplish the transformation. The transformation is everybody's job.

Clearly define top management's permanent commitment to ever improving quality and productivity, and their obligation to implement all of these principles.

Indeed, it is not enough that top management commit themselves for life to quality and productivity. They must know what it is that they are committed to—that is, what they must do.

Create a structure in top management that will push every day on the preceding 13 Points, and take action in order to accomplish the transformation. Support is not enough: action is required!





Benefits of Quality

- Higher customer satisfaction
- Reliable products/services
- Better efficiency of operations
- More productivity & profit
- Better morale of work force
- Less wastage costs
- Less Inspection costs

- Improved process
- More market share
- Spread of happiness & prosperity
- Better quality of life for all.





Seven Deadly Diseases

The "Seven Deadly Diseases" include:

- Lack of constancy of purpose
- Emphasis on short-term profits
- Evaluation by performance, merit rating, or annual review of performance
- Mobility of management
- Running a company on visible figures alone
- Excessive medical costs
- Excessive costs of warranty, fueled by lawyers who work for contingency fees





Barriers to TQM implementation

1. Lack of management commitment
2. Inability to change organizational culture
3. Improper planning
4. Lack of continuous training and education
5. Incompatible organizational structure and isolated individuals and departments
6. Ineffective measurement techniques and lack of access to data and results
7. Paying inadequate attention to internal and external customers
8. Inadequate use of empowerment and team work
9. Failure to continuously improve





THANK YOU!!!



